Banking Customer Churn Prediction

A comparative analysis of various supervised ML models
to find the most suitable one resilient to severe class imbalance

Harshank Nimonkar

Abstract

This report presents a comparative analysis of supervised machine learning models for customer churn prediction. The dataset is highly imbalanced (approximately 95% non-churn vs 5% churn), making accuracy an unreliable performance indicator. Accordingly, models are evaluated using imbalance-sensitive metrics such as Precision, Recall, F1-score, ROC-AUC, and Precision–Recall curves. Empirical results suggest that ensemble methods provide improved minority-class detection and stronger ranking performance relative to linear approaches, offering more effective identification of high-risk customers.

1 Problem Statement

The objective of this study is to predict whether a customer will churn based on historical behavioral and transactional features. Given the highly imbalanced nature of the dataset, the primary goal is to maximize minority-class detection performance while maintaining reasonable precision.

2 Dataset Characteristics

- Nigerian Banking Customer 360
- 22 original features and 2 targets (30d churn and 90d churn).
- For this analysis, we choose the 30 day churn target variable.
- 2 polytomous features with 3 and 37 categories; applied OneHot encoding.
- Total samples: 2M
- Majority class (Non-Churn): $\sim$95%
- Minority class (Churn): $\sim$5%
- Expanded feature space to 60 dimensions

A stratified train-test split (70%-30%) was used to preserve class distribution.

3 Evaluation Strategy

Due to class imbalance, the following metrics were prioritized:

- Minority Precision
- Minority Recall
- Minority F1-score
- ROC Curve and AUC
- Precision-Recall Curve and Average Precision (AP)

Accuracy is reported but not used for model selection.

4 Models Evaluated

The following models were compared:

1. Logistic Regression (Baseline) (LR)
2. Bernoulli Naive Bayes (BNB)
3. SGD Classifier (solver='hinge') (SGD)
4. Decision Tree (DT)
5. Random Forest (RF)
6. AdaBoost (AB)
7. Gradient Boosting (GB)

All models were trained under identical preprocessing and evaluation conditions.

Model	Accuracy	'1' Precision	'0' Precision	'1' Recall	'0' Recall	'1' F1	'0' F1
LR	0.9466	0.4101	0.9569	0.1556	0.9882	0.2256	0.9723
BNB	0.9502	0.6751	0.9503	0.0062	0.9998	0.0124	0.9744
SGD	0.9481	0.3593	0.9521	0.0492	0.9954	0.0866	0.9733
DT	0.9904	0.8950	0.9955	0.9153	0.9943	0.9051	0.9949
RF	0.9904	0.8951	0.9955	0.9154	0.9944	0.9051	0.9949
AB	0.9850	0.8939	0.9893	0.7950	0.9950	0.8416	0.9921
GB	0.9904	0.8950	0.9955	0.9154	0.9943	0.9051	0.9949

Table 1: Performance of different models evaluated across different metrics such as Precision, Recall and F1 scores for the majority and the minority classes. Class '1': Minority Class (Churn), Class '0': Majority Class (No Churn).

5 Baseline Performance

Logistic Regression was used as a baseline linear model.

- Accuracy $\approx 94.66\%$
- Minority Recall relatively low
- Minority F1 significantly lower than overall accuracy

This demonstrates that accuracy is misleading in imbalanced classification problems.

6 Tree-Based Models

The Decision Tree model showed substantial performance improvement:

- Significant increase in Minority Recall
- Strong improvement in Minority F1-score
- Overall accuracy $\approx 99\%$

This indicates nonlinear interactions among features.

7 Ensemble Methods

7.1 Random Forest

Random Forest reduced variance and stabilized tree performance:

- High minority detection capability
- Strong macro and weighted F1 scores

7.2 AdaBoost

AdaBoost improved Churn recall relative to baseline but was significantly less consistent than Random Forest and Gradient Boosting.

7.3 Gradient Boosting

Gradient Boosting demonstrated the strongest and most consistent performance:

- Highest Minority F1-score
- Highest Average Precision
- Strong ROC and PR curve dominance

8 Minority Class Analysis: Precision vs Recall

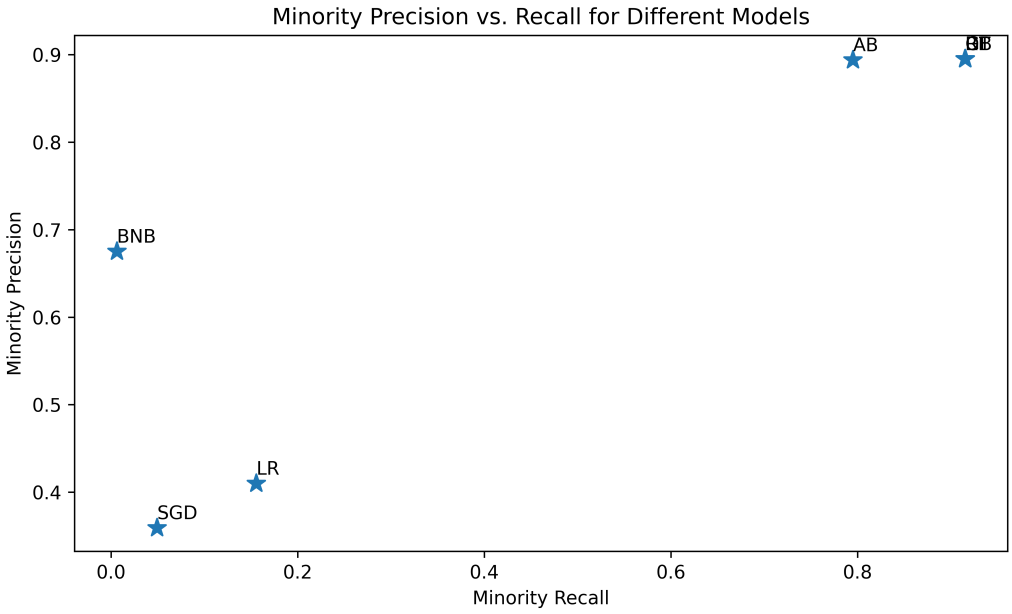

Figure 1: Minority Precision vs Minority Recall. Each star represents an individual model.

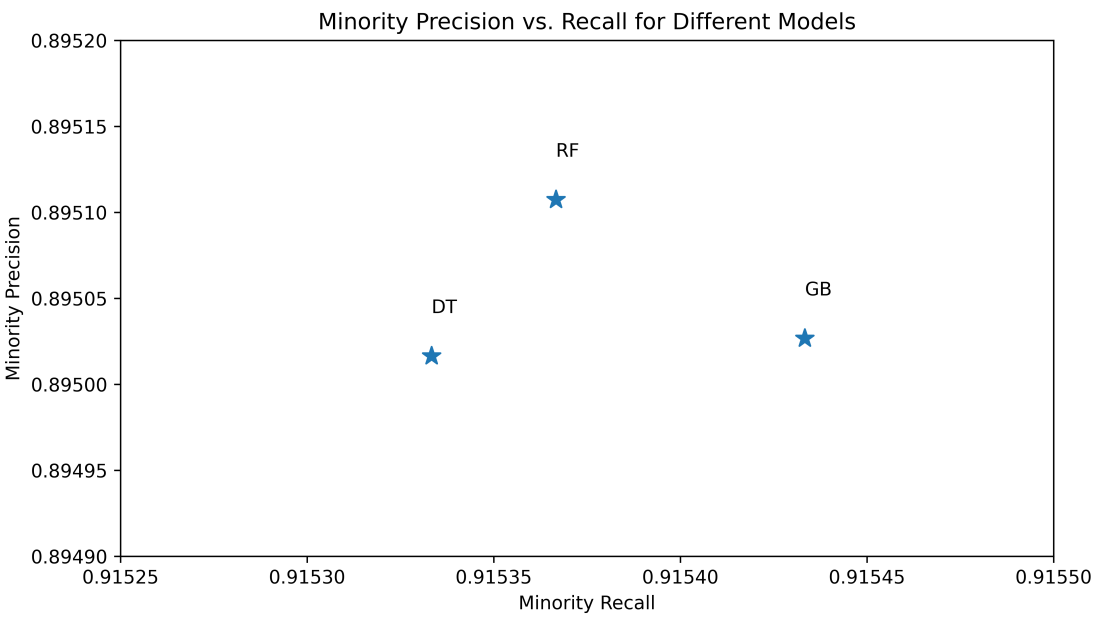

Figure 2: Minority Precision vs Minority Recall for top 3 performing models.

The precision and recall for the Churn class for different models shows a clear separation between the linear and non-linear models. The SGD (with 'hinge' loss) is basically a

Support Vector Classifier. The Bernoulli Naive Bayes Classifier, though a better guessor than LR and SGD, still fails to capture the non-linearities in the features. Tree-based and ensemble models succeed in this regard as evident from Fig. 1. We get a better look at the top right cluster in Fig. 2.
Looking at Fig. 2, the Precision-Recall tradeoff is apparent from the RF and GB positions. Depending on the business cost trade-offs, if retention outreach is inexpensive relative to churn loss, the model with higher recall will be preferred, i.e., Gradient Boosting. If outreach is expensive or limited in capacity, higher precision will be preferred, i.e., Random Forest.

9 ROC Curve Analysis

Figure 3 shows the ROC curve comparison across models. While all models exceed the random baseline, the ROC-AUC may overstate performance in this highly imbalanced setting (5% minority class), since the false positive rate is normalized by the large majority class. Consequently, even a considerable number of false positives can correspond to a relatively small false positive rate, making the curves appear optimistic.

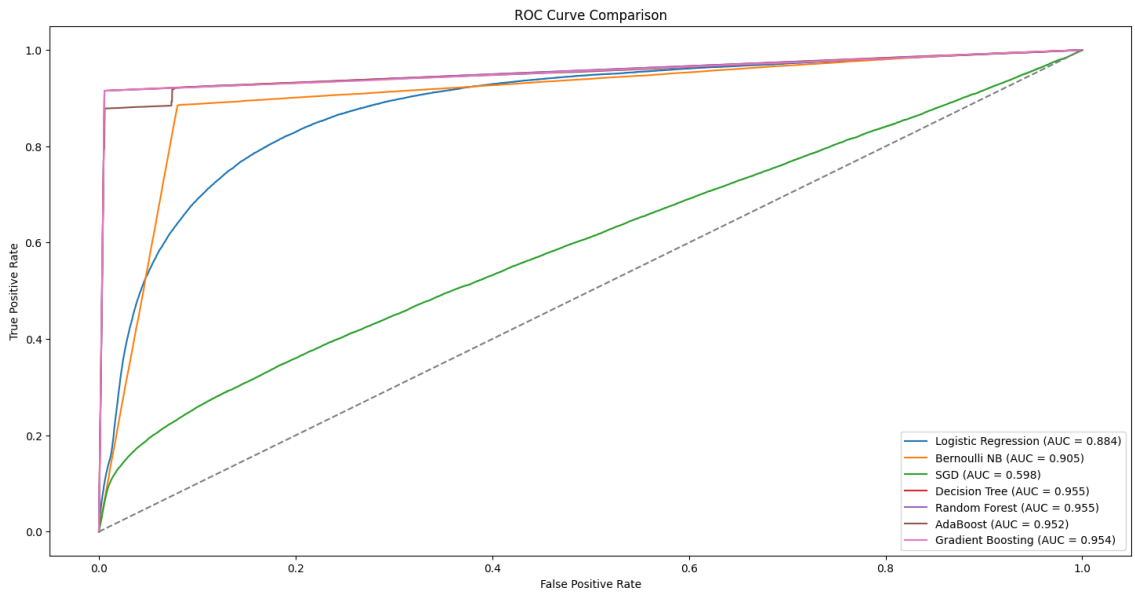

Figure 3: ROC Curve Comparison Across Models

10 Precision-Recall Curve Analysis

Figure 4 presents the Precision-Recall curve comparison. With a churn prevalence of approximately 5%, the expected precision under random prediction is ≈ 0.05, establishing a natural performance baseline. Ensemble models significantly exceed this baseline and outperform linear models in ranking minority-class instances, as reflected in improved ROC-AUC and precision-recall characteristics. This suggests superior ability to capture nonlinear interactions and better separate churners from non-churners in probability space, resulting in higher lift within top-ranked segments.

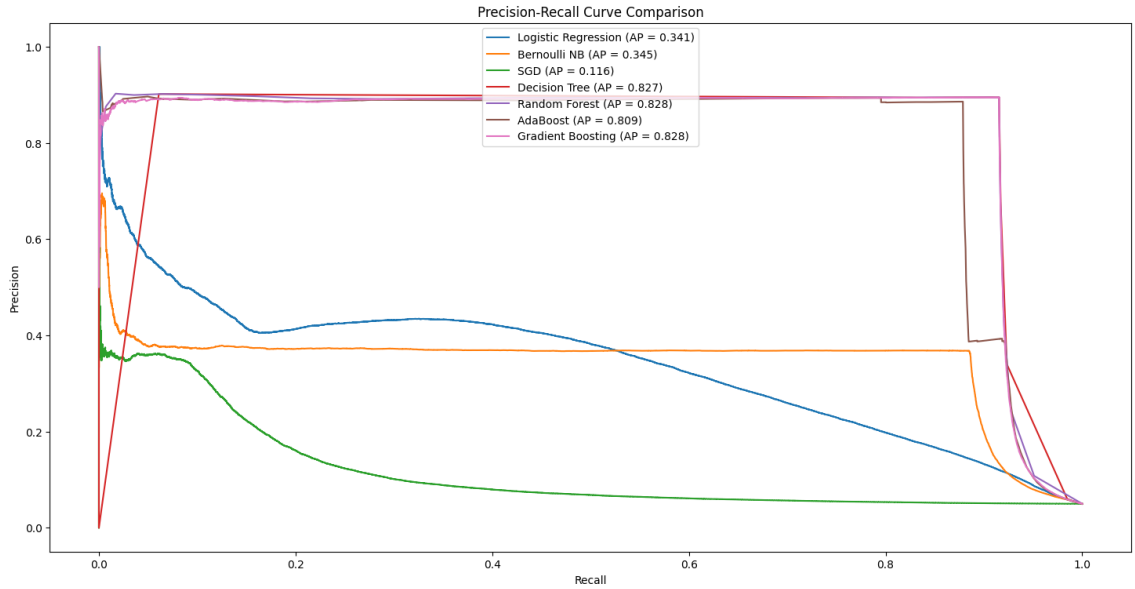

Figure 4: Precision-Recall Curve Comparison Across Models

11 Model Selection

Given the business objective of churn detection, minority recall and F1-score were prioritized. From the current analysis, Gradient Boosting is recommended closely followed by Random Forest due to:

- Superior minority detection
- Strong precision-recall tradeoff
- Robust overall performance

Further hyperparameter tuning may improve the precision–recall balance and support a more data-driven model selection decision.

12 Key Takeaways

1. Accuracy is insufficient in imbalanced classification.
2. Linear models underperform in nonlinear feature spaces.
3. Tree-based ensembles significantly improve minority detection.
4. Precision-Recall analysis is critical for rare-event prediction.

13 Future Work

Future improvements may include:

- Hyperparameter optimization
- Threshold tuning for business cost optimization
- Feature importance analysis and interpretation
- Cost-sensitive learning approaches